

AI Engineering

Working on AI Server

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Computer Science and Digital Communication



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there are 6 user accounts with user names aie1 ... aie6, one for each group

AI server:

```
USERNAME=aie[1-6]\\  
HOST=77.237.53.194\\  
password dahf7oeB
```

use secure shell ssh to connect to AI server

```
ssh -p 24 USERNAME@HOST
```

important to note: this ssh-service runs on port 24 (default: 22)

Working on server via ssh

use secure shell `ssh` to connect to server

```
ssh -p 24 USERNAME@HOST
```

important to note: this ssh-service runs on port 24 (default: 22)

SSH - password-less connection to the server from Unix-like OS (Mac/Linux)

automatic password-less authorization using public/private rsa key pair

if there is no public/private rsa key pair in \$HOME/.ssh generate one:

```
ssh-keygen
```

Copy user's public rsa key to the server for authentication by **appending** the rsa key to the existing `authorized_keys` file

```
cat $HOME/.ssh/id_rsa.pub |  
ssh -p 24 USERNAME@HOST "cat >> .ssh/authorized_keys"
```

SSH - password-less connection to the server from Windows OS

automatic password-less authorization using public/private rsa key pair

Use **Windows Powershell**:

if there is no public/private rsa key pair in \$env:USERPROFILE/.ssh generate one:

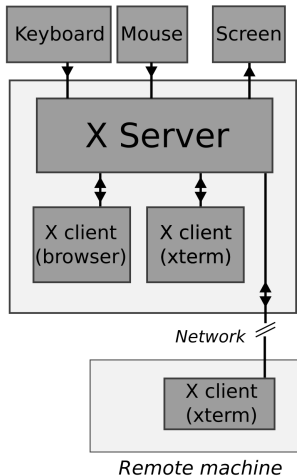
```
ssh-keygen
```

Copy user's public rsa key to the server for authentication by **appending** the rsa key to the existing `authorized_keys` file

```
type $env:USERPROFILE\.ssh\id_rsa.pub |  
ssh -X -Y -p 24 USERNAME@HOST "cat >> .ssh/authorized_keys"
```

Graphical User Interface on Unix/Linux/Mac OS

Unix-like operating systems use the X Window System to provide graphical user interfaces



display any graphical output on client side on Unix/Mac OS

remote/network client

use `-X` for untrusted X11 forwarding

use `-Y` for trusted X11 forwarding

```
ssh -X -Y -p 24 USERNAME@HOST
```

X Server Forwarding to non Unix Clients - Install X Server

on Windows clients install either:

- VcXsrv <https://sourceforge.net/projects/vcxsrv/>
- Xming <https://sourceforge.net/projects/xming/>
- MobaXterm <https://mobatek.net/>

on Mac OS clients install

<https://xquartz.org/>

<https://docs.cse.lehigh.edu/xforwarding/xforwarding-mac/>

Windows - Server file exchange

copy from local host to remote host (use hostname or IP)

```
scp SOURCE USERNAME@HOST:PATH/TARGET
```

copy from remote host (use hostname or IP) to local host

```
scp USERNAME@HOST:PATH/SOURCE TARGET
```

alternatively use WINSCP on Windows OS

<https://winscp.net/eng/download.php>

Install and activate virtual environment

Unix/MaxOS

```
# install environment, ensure python 3.10 version
python3 -m venv .venv
```

```
# activate environment
source .venv/bin/activate
```

Windows

```
# install environment, ensure python 3.10 version
py -m venv .venv
```

```
# activate environment
.venv\Scripts\activate
```

Note: `.env` denotes the name of the directory in which the environment is installed to. Repeating the install instruction overwrites/deletes all previously installed python packages and configurations

Install required python packages in virtual environment

```
# install all required packages
pip install tensorflow==2.20.0
pip install keras==3.12.0
pip install matplotlib
pip install numpy
pip install scikit-learn
...
```

Unix

```
# matplotlib and others plotting packages require pyqt5
pip install pyqt5
```

Important to note: only install packages if the virtual environment is active !

AI Server - Collaborative Environment

Goal: minimize resource consumption while maintaining efficient experimentation - ensuring that the shared infrastructure remains accessible and fair for all collaborators

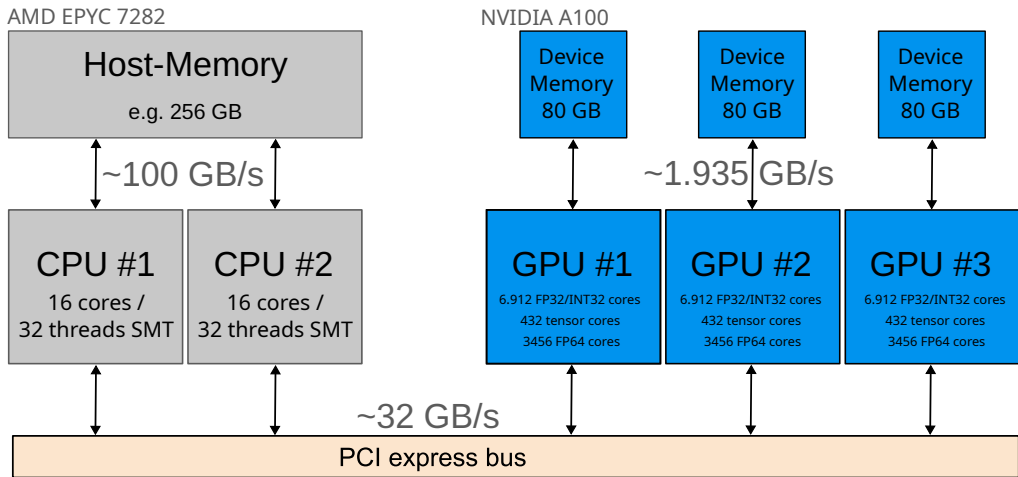
- Resources are limited:
 - CPU cores, GPU compute capacity
 - system memory (RAM)
 - disk storage
- Resources are shared across:
 - Multiple users
 - Courseworks and project groups
- Goal:
 - Minimize resource usage
 - Ensure fair and efficient access for everyone: overconsumption negatively impacts work of others and may freeze OS

→ important to manage shared computational resources carefully so that all users and course projects can run efficiently

AI Server - Collaborative Environment - Best Practises

1. Use GPUs only when necessary
Run small experiments or debugging on CPU before moving to GPU training
2. Use efficient and appropriate datatypes
(e.g., float32 or mixed precision float16)
3. Optimize batch sizes and model complexity
Choose reasonable batch sizes and avoid unnecessarily large models that consume excessive GPU memory
4. Release GPU memory when finished
Properly terminate sessions or scripts to free GPU resources
5. Schedule long training jobs appropriately
Avoid running heavy jobs during peak usage times if possible
6. Monitor resource usage
Regularly check GPU/CPU utilization and memory consumption (e.g., using system monitoring tools, e.g. nvidia-smi, nvidia-smi, nvidia-smi)
7. Manage storage efficiently
Remove unnecessary checkpoints, logs, and intermediate datasets
8. Use efficient data pipelines
Optimize TensorFlow data loading (e.g., batching, caching, prefetching) to reduce unnecessary computation

CPU versus GPU: HCW CSDC AI Server



Server GPU information

```
mlt_ml1@ai-server:~$ nvidia-smi
```

NVIDIA-SMI 565.57.01					Driver Version: 565.57.01			CUDA Version: 12.7	

GPU	Name		Persistence-M			Bus-Id	Disp.A	Volatile Uncorr. ECC	
Fan	Temp	Perf	Pwr:Usage/Cap			Memory-Usage		GPU-Util	Compute M.
								MIG M.	
=====									
0	NVIDIA	A100	80GB	PCIe	Off	00000000:41:00.0 Off		0	
N/A	37C	P0	63W / 300W			3161MiB / 81920MiB		0%	Default
								Disabled	

1	NVIDIA	A100	80GB	PCIe	Off	00000000:61:00.0 Off		0	
N/A	33C	P0	42W / 300W			17MiB / 81920MiB		0%	Default
								Disabled	

2	NVIDIA	A100	80GB	PCIe	Off	00000000:A1:00.0 Off		0	
N/A	33C	P0	43W / 300W			17MiB / 81920MiB		0%	Default
								Disabled	

Server GPU information

Processes:						
GPU	GI	CI	PID	Type	Process name	GPU Memory Usage
	ID	ID				
0	N/A	N/A	2111	G	/usr/lib/xorg/Xorg	4MiB
0	N/A	N/A	1689315	C	venv/bin/python	3138MiB
1	N/A	N/A	2111	G	/usr/lib/xorg/Xorg	4MiB
2	N/A	N/A	2111	G	/usr/lib/xorg/Xorg	4MiB

... running processes using GPU

to monitor the GPU usage continuously ...

```
mlt_ml1@ai-server:~$ watch -n0.1 nvidia-smi
mlt_ml1@ai-server:~$ nvidia-smi
```

try it out ...

Server GPU usage - minimize memory usage

constrain the allocation of GPU memory

```
import tensorflow
...

gpus = tensorflow.config.experimental.list_physical_devices('GPU')
if gpus:
    try:
        for gpu in gpus:
            tensorflow.config.experimental.set_memory_growth(gpu, True)
        print("Memory growth set for GPUs")
    except RuntimeError as e:
        print(e)
```

Comparison of Dtypes for Deep Learning (A100) (1)

■ Sign (S) ■ Exponent ■ Mantissa (Fraction)

Float32 (Single Precision)



Float16 (Half Precision)



BFloat16 (Brain Float)



Dtype	Precision (Mantissa)	Dynamic Range (Exponent)	Application
Float32	High	Standard (10^{-38} to 10^{38})	Used for Master Weights
Mixed FP16	Medium	Low (10^{-5} to $65k$)	Requires Loss Scaling to avoid zero-gradient ¹
Mixed BF16	Low	High (10^{-38} to 10^{38})	Native on A100; no loss scaling needed

¹ Loss scaling multiplies the loss by S to avoid gradient underflow in float16; gradients are rescaled by $1/S$ before the update.

Comparison of GPU Dtypes for Deep Learning - float32

dtype=float32 nvidia-smi

GPU						Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr.	ECC
Fan	Temp	Perf	Pwr:Usage/Cap				Memory-Usage		GPU-Util	Compute	M.
									MIG M.		
=====											
0	NVIDIA	A100 80GB PCIe	Off			00000000:41:00.0	Off		0		
N/A	64C	P0	189W / 300W			4625MiB / 81920MiB		74%	Default		

Processes:										
GPU	GI	CI	PID	Type	Process name				GPU Memory	
	ID	ID							Usage	
=====										
0	N/A	N/A	1913185	C	python3.13				4594MiB	

Comparison of GPU Dtypes for Deep Learning - mixed-float16

```
dtype=mixed_float16 nvidia-smi
```

GPU Name					Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr. ECC
Fan	Temp	Perf	Pwr:Usage/Cap			Memory-Usage	GPU-Util	Compute M.	
								MIG M.	
=====									
0	NVIDIA	A100	80GB	PCIe	Off	00000000:41:00.0	Off		0
N/A	59C	P0	127W	/ 300W		2529MiB / 81920MiB	66%	Default	

Processes:						
GPU	GI	CI	PID	Type	Process name	GPU Memory
	ID	ID				Usage
=====						
0	N/A	N/A	1906249	C	python3.13	2490MiB

Comparison of GPU Dtypes for Deep Learning - mixed-bfloat16

dtype=mixed_bfloat16 nvidia-smi

GPU						Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr.	ECC
Name									Memory-Usage	GPU-Util	Compute M.
Fan Temp Perf						Pwr:Usage/Cap					MIG M.
0	NVIDIA	A100	80GB	PCIe		Off	00000000:41:00.0	Off			0
N/A	57C	P0				196W / 300W	2529MiB / 81920MiB		71%		Default

Processes:											
GPU	GI	CI	PID	Type	Process name					GPU Memory	
	ID	ID								Usage	
0	N/A	N/A	1900873	C	python3.13					2490MiB	

Comparison of GPU Dtypes for Deep Learning - throughput

Example: A Convolutional Residual-Encoder-Classification ANN

Dtype	Training time [s]	Throughput [<i>samples/s</i>]	Memory [<i>MB</i>]
float32	380	270	4594
mixed_float16	192	533	2490
mixed_bfloat16	254	403	2490

roughly half the memory used and roughly $2\times$ faster for mixed_float16, mixed_bfloat16,

Server CPU/GPU usage - setting up the device and datatype (1)

```
device = ...

if device.lower() == "cpu":
    tensorflow.config.set_visible_devices([], "GPU")
    mixed_precision.set_global_policy("float32")
    print("Using CPU only.")
    return

self._gpus = tensorflow.config.list_physical_devices('GPU')

if not self._gpus:
    print("Non GPUs found, falling back to CPU")
    mixed_precision.set_global_policy("float32")
    return

try:
    selected_gpus = [self._gpus[i] for i in gpu_ids]
except IndexError:
    raise ValueError(f"Requested GPU ids {gpu_ids}, but only {len(self._gpus)} \\  
GPUs available")
...

```


Server CPU/GPU usage - setting up the device and datatype (2)

```
...
tensorflow.config.set_visible_devices(selected_gpus, 'GPU')
for gpu in selected_gpus:
    if memory_limit is not None:
        tensorflow.config.set_logical_device_configuration(gpu,
            [tensorflow.config.LogicalDeviceConfiguration( memory_limit=memory_limit_mb)])
    else:
        tensorflow.config.experimental.set_memory_growth(gpu, True)

print(f"Using GPU(s): {selected_gpus}")

precision_policy = PrecisionPolicy.from_string(hardware.get("precision", "float32"))

mixed_precision.set_global_policy(precision_policy.value)
print("Precision policy:", mixed_precision.global_policy())
```

Server GPU usage - use dedicated GPU

select specific GPU to be used while computing with tensorflow

```
import tensorflow
...
if len(gpus) <= args.gpu:
    print("Number of available GPU {} smaller than requested number {}, \
        set number used to number available\n".format(len(gpus), args.gpu))
    num_gpus = len(gpus)
else:
    try:
        tensorflow.config.experimental.set_visible_devices(gpus[args.gpu], 'GPU')
    except RuntimeError as e:
        print(e)
```

Unix/Linux tools environment to work with

familiarize with working in the Linux command line

<https://ubuntu.com/tutorials/command-line-for-beginners#1-overview>

<https://www.freecodecamp.org/news/the-linux-commands-handbook/>

Bash shell

<https://www.geeksforgeeks.org/bash-scripting-introduction-to-bash-and-bash-scripting/>

Text processing tools: `wc`, `grep`, `head`, `tail`, `more`, `cut`, `awk`, `sed`, `sort`, `tr`, ...

Process handling tools: `ps`, `kill`, `top`, ...

Editors: in command window nano, vi, or on X Server xedit, gedit

Environment variables: HOME, PATH, LD_LIBRARY_PATH, PYTHONPATH

Configuration of account in \$HOME/.bashrc